Ericsson PE Node WHI-PE-14 Hardware/Environmental Fault Troubleshooting and Resolution SOP

Node: WHI-PE-14 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_2557

Description

This Standard Operating Procedure outlines the troubleshooting and resolution steps for a Hardware/Environmental Fault error in the PE node (WHI-PE-14) at the Whitefield East POP. The error is characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss, as indicated by Alarm Code ALARM_2557, with a severity of Warning. The Ericsson vendor-specific CLI dialect is employed throughout this SOP. The underlying protocol mechanics responsible for this failure domain involve the PE node, which is a critical component of the Ericsson routing architecture. The PE node is responsible for processing and forwarding traffic between the IP backbone and the access network. The ASIC memory parity errors and line card crashes can be attributed to hardware failures or environmental stressors such as overheating, power surges, or electrical interference. The loss of power supply redundancy can be due to a faulty or malfunctioning power supply unit (PSU). This SOP aims to identify the root cause of the failure and implement corrective measures to restore normal operation.

Pre-requisite Checks

- check: Verify the site location and ensure access to the Whitefield East POP for remote hands
- check: Check the redundant link status and ensure it is active
- check: Review the alarm history and logs for similar issues

Potential Root Causes

Cause	Probability
ASIC memory parity error due to hardware failure	High
Line card crash due to software or hardware issue	Medium
Power supply redundancy loss due to faulty PSU	Low

Diagnostic Commands & Expected Outputs

```
Command: show system redundancy, Expected Output: System redundancy status: Active, Description: This command verifies the system redundancy status, ensuring that the redundant link is active and functional.

Command: show memory parity errors, Expected Output: Memory parity errors: 10, Description: This command displays the number of memory parity errors, which can indicate hardware failures or environmental
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stressors.

Command: show line card status, Expected Output: Line card status:
Online, Description: This command checks the status of the line card,
ensuring it is online and functional.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC or line card, Command: replace asic slot 1, Description: This step involves replacing the faulty ASIC or line card with a new one to resolve the hardware failure.

Step 2: Step: Replace faulty PSU, Command: replace power-supply 1, Description: This step involves replacing the faulty PSU with a new one to resolve the power supply redundancy loss.

Step 3: Step: Perform a system reboot, Command: system reboot, Description: This step involves rebooting the system to ensure that the changes take effect and the system operates normally.

Post-Check Verification

```
Command: show system status, Expected Output: System status: Normal,
Description: This command verifies the system status, ensuring it is
normal and functional after the resolution steps.

Command: show memory parity errors, Expected Output: Memory parity
errors: 0, Description: This command checks the number of memory
parity errors, ensuring it is 0 to confirm the resolution of the
hardware failure.

Command: show line card status, Expected Output: Line card status:
Online, Description: This command checks the status of the line card,
ensuring it is online and functional after the resolution steps.
```

Roll-back Procedures

- Step: Restore original components, Command: undo replace asic slot 1, Description: This step involves restoring the original ASIC or line card to revert the changes made during the resolution steps.

- Step: Restore original PSU, Command: undo replace power-supply 1, Description: This step involves restoring the original PSU to revert the changes made during the resolution steps.